

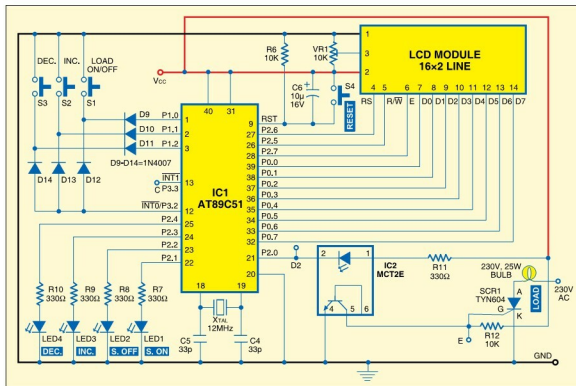


Fig. 2: Circuit diagram of phase angle control of SCR using AT89C51

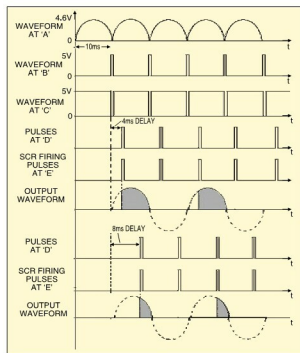


Fig. 3: Waveforms observed at various points in Fig. 1 and Fig. 2 and SCR output waveforms

module. Port pins P2.6, P2.5 and P2.7 of the microcontroller control the registers select (RS), read/write (R/W) and enable (E) input pin of the LCD

respectively. The role of different switches is shown in Table I.

The output of the zero-crossing detector from 'C' is fed to the external

TABLE I
Role of
Different Switches

Switch	Function
S1	To switch SCR on/off
S2	Increase delay by 1 sec
S3	Decrease delay by 1 sec

module, respectively. Preset VR1 is used for controlling the contrast of the LCD module. Push-to-on switches S1, S2 and S3 are connected with the pins P1.0, P1.1 and P1.2 through diodes D9, D10 and D11, respectively. External interrupt pin INT0 (P3.2) of the microcontroller is connected to S1, S2 and S3 through D12, D13 and D14, respectively.

The operation

The complete operation can be well understood with the help of waveforms in Fig. 3.

1. The waveform at point 'A' is a fully rectified wave that is fed to the base of T1.

2. When the base voltage falls below 0.7V, transistor T1 is switched off, pulling the output higher. This results in a very short positive pulse, which is available at the collector, (at point 'B') as shown in the second waveform.

3. As this positive pulse is inverted by transistor T2, it produces one negative pulse of the same width at 'C'. This is shown as the third

interrupt pin INT1 (P3.3) of the microcontroller.

Port pin P2.0 is connected with pin 2 of the opto-coupler (MCT2E). The output pin 5 of MCT2E is used for triggering the gate of SCR TYN604. The anode of SCR is connected to the load (bulb) with the 230V AC supply.

A 12MHz crystal along with capacitors C5 and